

①.

$$\operatorname{tg}^5 x - 9 \operatorname{tg} x = 0$$

$$\operatorname{tg} x [\operatorname{tg}^4 x - 9] = 0$$

$$\operatorname{tg} x [(\operatorname{tg}^2 x - 3)(\operatorname{tg}^2 x + 3)] = 0$$

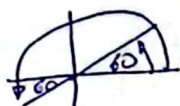
$$\operatorname{tg} x (\operatorname{tg} x - \sqrt{3})(\operatorname{tg} x + \sqrt{3})(\operatorname{tg}^2 x + 3) = 0$$

donde

$$\operatorname{tg} x = 0 \Rightarrow x = 0, \pi, 2\pi$$

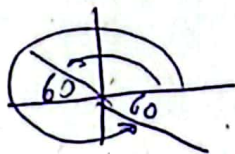
$$\operatorname{tg} x - \sqrt{3} = 0 \Rightarrow \operatorname{tg} x = \sqrt{3} \Rightarrow x = 60^\circ = \frac{\pi}{3}$$

$$x = 240^\circ = \frac{4\pi}{3}$$



$$\operatorname{tg} x + \sqrt{3} = 0 \Rightarrow \operatorname{tg} x = -\sqrt{3} \Rightarrow x = 120^\circ = \frac{2\pi}{3}$$

$$x = 300^\circ = \frac{5\pi}{3}$$

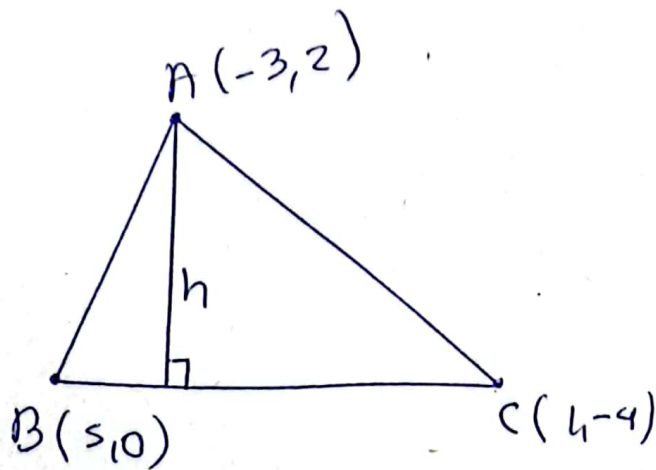


$$\operatorname{tg}^2 x + 3 = 0 \Rightarrow \operatorname{tg} x = \sqrt{-3} \rightarrow \text{No solution}$$

la suma de soluciones es:

$$\text{Suma} = (0 + \pi + 2\pi) + \left(\frac{\pi}{3} + \frac{4\pi}{3}\right) + \left(\frac{2\pi}{3} + \frac{5\pi}{3}\right)$$

$$= 7\pi$$



La recta BC, está dada por

$$y - y_0 = m(x - x_0)$$

$$y - 0 = \frac{-4 - 0}{1 - 5}(x - 5)$$

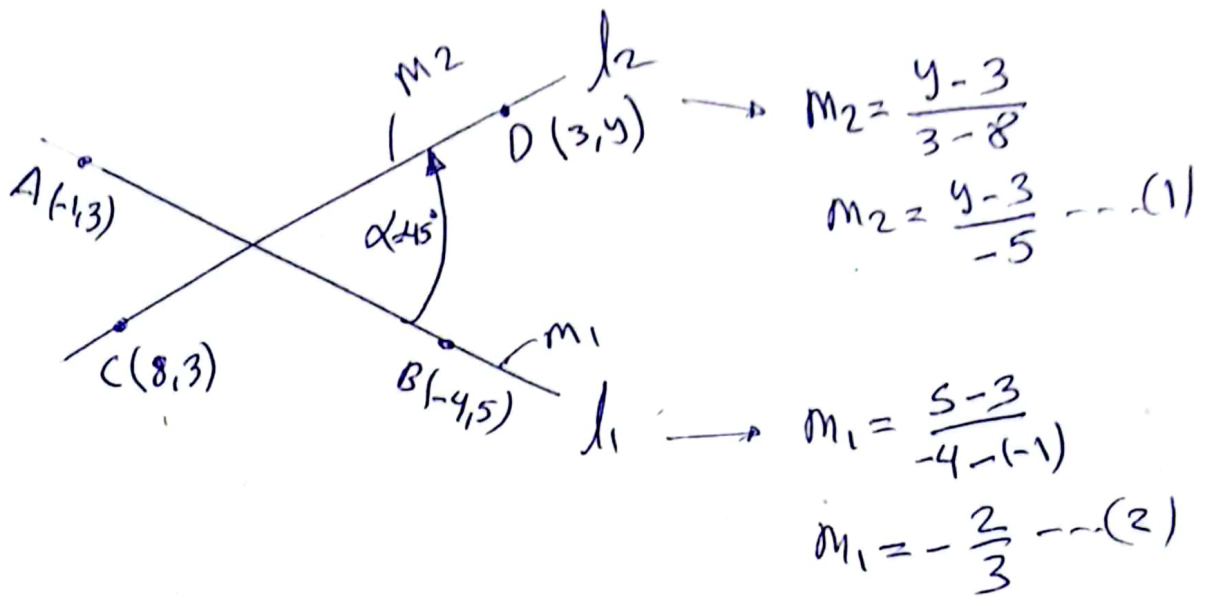
$$y = x - 5 \Rightarrow x - y - 5 = 0$$

La altura del punto A a la Recta BC, corresponde la distancia del punto A, a la recta BC

$$h = d = \frac{|(-3) - (2) - 5|}{\sqrt{1^2 + (-1)^2}} = \frac{10}{\sqrt{2}} \cdot \left(\frac{\sqrt{2}}{\sqrt{2}}\right)$$

$$\underline{h = 5\sqrt{2}}$$

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luego

$$\tan \alpha = \frac{m_2 - m_1}{1 + m_2 m_1}$$

$$\tan 45 = \frac{m_2 - (-\frac{2}{3})}{1 + m_2 \cdot (-\frac{2}{3})}$$

$$1 = \frac{\frac{3m_2 + 2}{3}}{\frac{3 - 2m_2}{3}}$$

$$3 - 2m_2 = 3m_2 + 2 \Rightarrow m_2 = \frac{1}{5}$$

En (1) se tiene:

$$m_2 = \frac{y-3}{-5}$$

$$\frac{1}{5} = \frac{y-3}{-5} \Rightarrow \boxed{y=2}$$

(4)

$$E = \frac{2 - \csc^2 x}{\tan x - 1} - \csc^2 x + 1$$

$$E = \frac{2 - (1 + \cot^2 x)}{\tan x - 1} - (1 + \cot^2 x + 1)$$

$$= \frac{1 - \cot^2 x}{\tan x - 1} - \cot^2 x$$

$$= \frac{1 - \frac{1}{\tan^2 x}}{\tan x - 1} - \frac{1}{\tan^2 x}$$

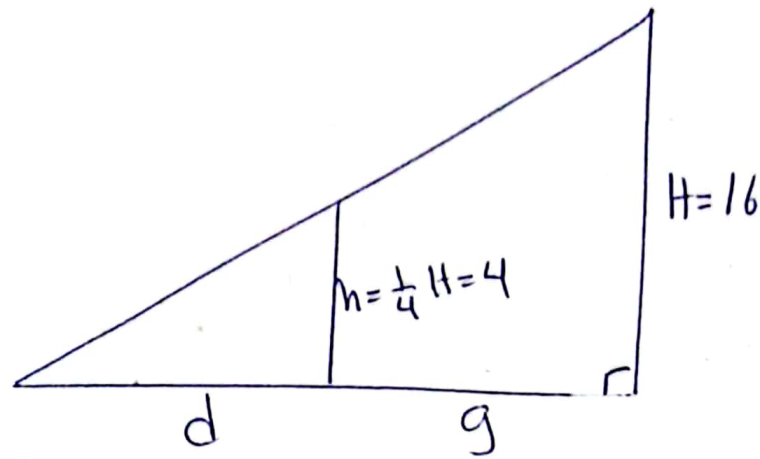
$$= \frac{\frac{\tan^2 x - 1}{\tan^2 x}}{\tan x - 1} - \frac{1}{\tan^2 x}$$

$$= \frac{(\cancel{\tan x - 1})(\tan x + 1)}{\tan^2 x (\cancel{\tan x - 1})} - \frac{1}{\tan^2 x}$$

$$= \frac{\tan x + 1 - 1}{\tan^2 x}$$

$$= \frac{1}{\tan x} = \cot x \rightarrow$$

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Los triángulos son proporcionales, entonces:

$$\frac{d+g}{d} = \frac{16}{4}$$

$$d+g = 4d \Rightarrow \boxed{d=3}$$

luego $\tan \theta = \frac{4}{3} \Rightarrow \theta = \tan^{-1}\left(\frac{4}{3}\right)$

$$\theta = 53^\circ 7' 48.37''$$